

Eco-Loop Building Agents: quantitative savings

EnergyPlus closed loop, 168 simulated hours · baseline baseLine vs agent agent_11a · energy from EnergyPlus eplusmtr.csv

TOTAL ELECTRICITY SAVED

+8.8%

1100 → 1004 kWh

HVAC ELECTRICITY SAVED

+23.3%

414 → 317 kWh

COMFORT IN BAND

93.1%

+12.4 pts vs baseline's 80.7%

CO₂ AVOIDED

62 kg

664 → 602 kg

COST SAVED

₹594.49

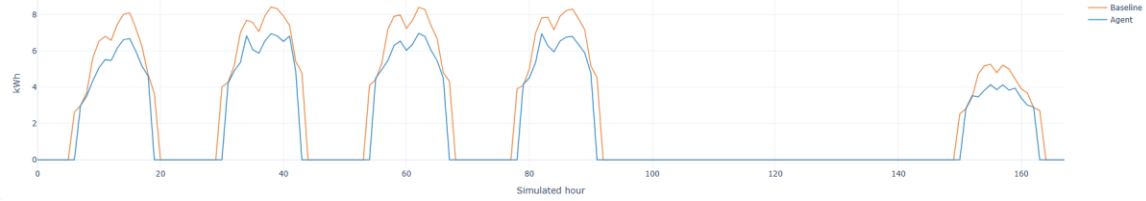
at ₹5/kWh tariff, over 168 simulated hours

PEAK DEMAND

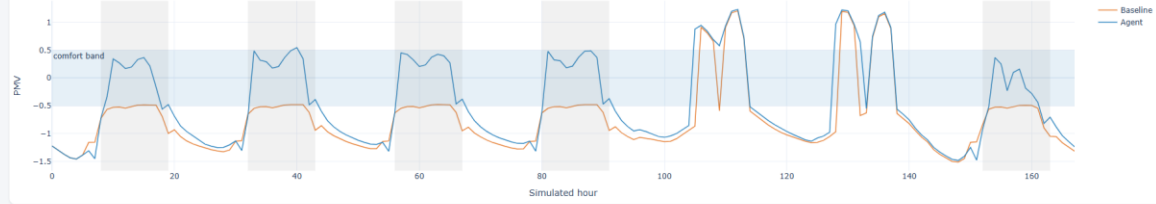
19.0 kW

baseline 19.9 kW

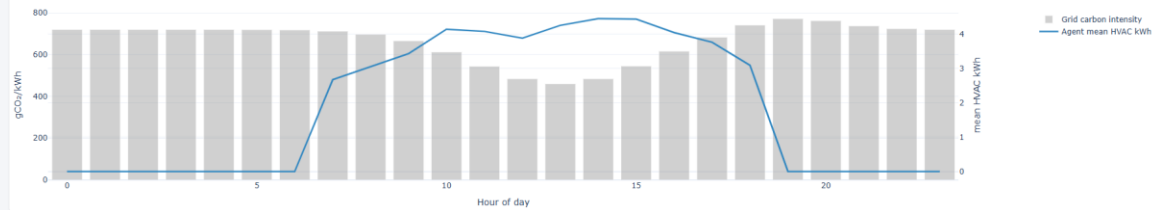
HVAC electricity, hour by hour (EnergyPlus eplusmtr.csv)



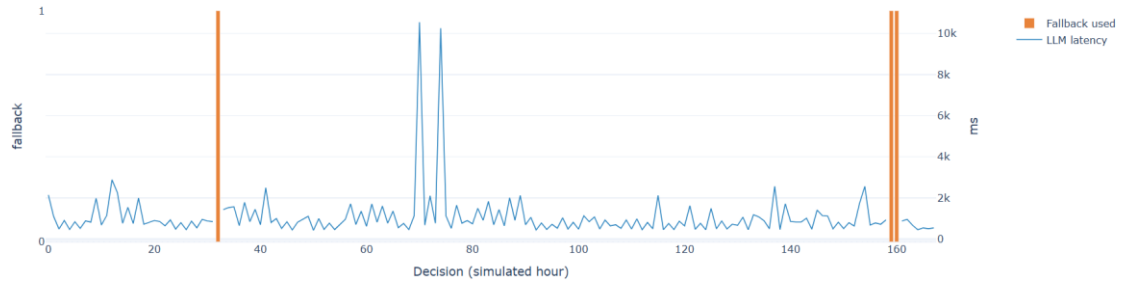
Worst-zone PMV vs the comfort band (shaded columns = occupied hours)



Carbon-aware load shifting: when the agent spends energy



Decision reliability: fallback usage and LLM latency per hour



14-day LLM endurance run: 336/336 decisions, 0 fallback, 92.9% comfort in-band, latency p50 693 ms / p95 1641 ms

168 decisions · 3 fallback · 0 retried · latency p50 816 ms / p95 2120 ms